Forecasting Bitcoin Prices and Analyzing Geopolitical Risks: A Residual Analysis Approach using LSTM

Technical Report 01 (TR-01):

Preliminary Study on Bitcoin Trend Classification using Random Forest and Sentiment Analysis Proxies

1. Abstract

This technical report documents the initial technical kick-off of 'Project Geo-BTC: Deep Learning & Macro-Inference', aimed at establishing a robust Machine Learning pipeline and investigating non-financial data proxies. The foundational pipeline utilized the *yfinance* library for historical daily Bitcoin market data. Regarding feature engineering, we implemented a novel, low-cost sentiment analysis approach by mining edit frequencies on the 'Bitcoin' Wikipedia page using the *mwclient* library, serving as a quantitative proxy for public sentiment and market attention. A Random Forest Classifier was trained for binary classification of daily price direction (Up/Down). While this architecture served as an effective technical baseline for team onboarding and initial data exploration, the achieved Precision Score demonstrated limitations inherent to the model's independence assumption. Empirical results validate the strategic necessity of transitioning to Deep Learning architectures, specifically LSTM Networks, to capture complex temporal dependencies and move from classification to high-fidelity price regression in the subsequent quantitative phase (Phase 1).

2. Introduction

This technical report outlines the preliminary phase of the 'Project Geo-BTC'. Before implementing complex Deep Learning architectures, the squad aimed to establish a unified Machine Learning workflow. The goal of this phase is twofold: (1) to technically onboard the team regarding Python data engineering pipelines, and (2) to test a baseline hypothesis that public interest (measured via Wikipedia edits) correlates with Bitcoin price volatility. This report details the data acquisition, the choice of a Random Forest baseline, and the limitations that motivate the transition to LSTM networks.

2. Objective

The primary objective of this study was to architect and validate a fundamental end-to-end Machine Learning pipeline. This phase aims to operationalize data acquisition, preprocessing, and feature engineering modules, establishing a reproducible workflow for the research team. Additionally, the study sought to define a performance baseline using a deterministic model (Random Forest) to rigorously benchmark the subsequent Deep Learning implementations within the Python/Jupyter environment.

2.1 Scope

This report covers the initial analysis of daily Bitcoin price movements (Up/Down classification) combined with a metric of public interest (Wikipedia edit frequency). This exploration served to test the feasibility of integrating non-traditional features and defining a minimum viable data pipeline before the definitive Phase 1 implementation

3. Data Engineering

This section details the process of acquisition, processing, and feature engineering for the preliminary classification phase.

3.1 Financial Data Acquisition

Historical market data (Open, High, Low, Close, Volume - OHLCV) was programmatically acquired using an *yfinance* API.

Granularity Rationale: The selection of daily granularity for this initial study was a temporary, strategic measure. Its purpose was to simplify the data preparation workload and expedite the establishment of the baseline model. However, for the definitive quantitative phase (Phase 1), the model will require the use of the minute-level dataset (bitstamp_cleaned) to fully capture the intra-day temporal dependencies essential for high-fidelity LSTM forecasting.

3.2 External Feature Proxy: Public Sentiment

A novel approach for incorporating public sentiment was tested. The underlying hypothesis posits that significant increases in public interest, quantified by activity on the 'Bitcoin' Wikipedia page, function as a leading indicator of increased market attention and potential volatility.

The *mwclient* library was utilized to interface with the *MediaWiki* API, enabling the extraction and daily aggregation of unique page edit counts. This methodology provided a low-latency, quantifiable proxy for external public engagement, which was subsequently synchronized with the financial time series.

3.3 Feature Engineering: Rolling Window Application

The sentiment proxy is derived from the edit frequency of the Wikipedia article. Let E_t represent the raw count of edits on day t. To mitigate high-frequency noise and capture the underlying trend of public attention, we apply a rolling window smoothing function over a period w. The smoothed feature S_t is calculated as:

$$S_t = \frac{1}{w} \sum_{i=0}^{w-1} E_{t-i}$$

This transformation allows the model to learn from the accumulated interest momentum rather than erratic daily spikes.

4. Methodology

The baseline classification is performed using Random Forest, an ensemble method aggregating K decision trees. Let $h_k(x)$ denote the prediction of the $k\text{-th}$ individual tree for an input vector x. The final class prediction $\hat{Y}$ is determined by majority voting:

$$\hat{Y} = \arg max \sum_{k=1}^{k} \mathbb{I}(h_k(x) = c)$$

Where $\mathbb{I}(\cdot)$ is the indicator function. This ensemble approach reduces variance compared to single decision trees, providing a more stable baseline for non-stationary financial data.

4.1 Target Definition: Binary Classification

To frame the market prediction as a supervised classification task, we define the target variable y_t representing the directional movement of Bitcoin's closing price P_t at time t relative to the previous time step t-1. The binary target is formalized as:

$$y_t = \begin{cases} 1, & P_t > P_{t-1} \ (Up) \\ 0, & P_t \leq P_{t-1} \ (Down) \end{cases}$$

Where P_t denotes the closing price derived from the OHLCV (Open, High, Low, Close, Volume) vector.

4.2 Evaluation Metric: Precision Score

Given the financial implications of false positives (predicting an upward trend when the market declines), we prioritized the Precision metric over global Accuracy. Precision quantifies the reliability of positive predictions $(y=1)$ and is defined as:

$$Precision = \frac{TP}{TP + FP}$$

Where TP (True Positives) represents correctly predicted price increases, and FP (False Positives) denotes instances where the model incorrectly signaled a buying opportunity during a market downturn.

5. Results and Discussion

The model achieved a Precision Score of ≃59%, wich suggests that while it has a moderate ability to correctly identify upward movements, the overall accuracy remains below the required threshold for operational use.

However, reliance solely on the Precision Score (approx. 59%) provides an incomplete assessment of model viability for financial operations. In high-frequency or volatile markets, directional accuracy does not strictly equate to profitability, as it fails to account for the magnitude of price movements. Furthermore, the Random Forest algorithm fundamentally treats daily observations as independent events. This assumption neglects the intrinsic sequential dependencies and long-term trends of time-series data. Consequently, the model exhibits reactive behavior rather than a predictive one, often identifying a trend only after it has already established itself. This limitation confirms the hypothesis that non-recurrent architectures are insufficient for capturing the complex, non-linear dynamics of Bitcoin pricing, necessitating the transition to Long Short-Term Memory (LSTM) networks in the subsequent phase.

Figure 1.

Chart showing the evolution of Bitcoin's price over time and an analysis of market sentiment over time, superimposed.

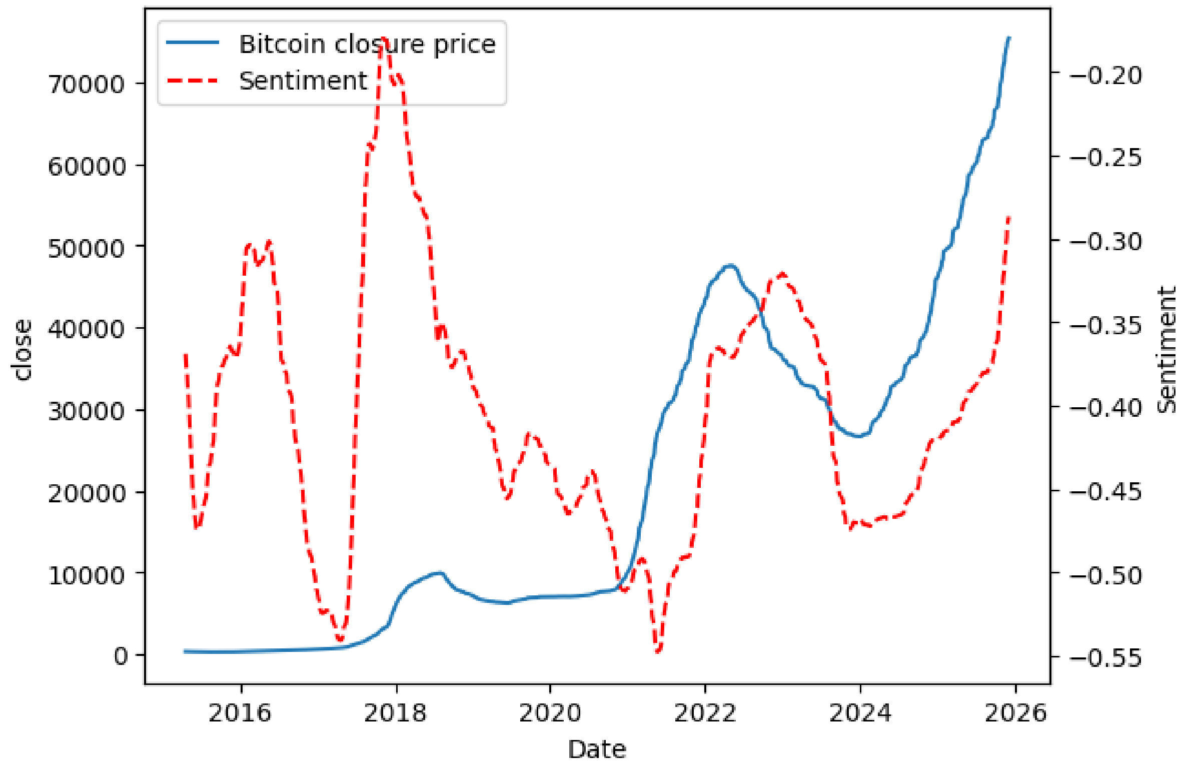

Figure 1 illustrates the raw edit counts versus the moving average. The visualization confirms that the 'Rolling Window' technique was an effective preprocessing step (feature engineering), successfully filtering daily noise to reveal underlying trends in public interest.

Figure 2.

Comparative analysis of the 30-Day Moving Average (MA) for Actual vs. Predicted values. The divergence highlights the model's latency in responding to sharp market corrections.

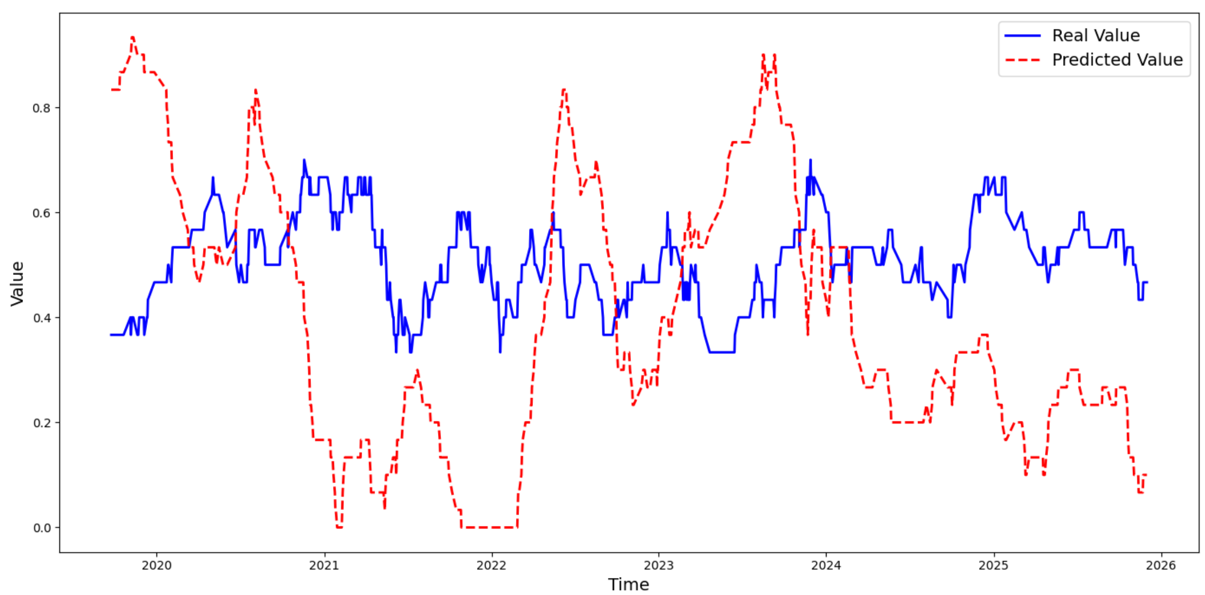

Figure 2 visualizes the smoothing effect of the predictions compared to the actual market trajectory. As observed, the Random Forest model exhibits a significant 'lag effect.' While it successfully captures the general long-term trend, it fails to anticipate rapid volatility spikes (both bullish and bearish). The predicted curve (Orange) consistently trails the actual price curve (Blue), indicating that the model is heavily weighted by immediate past values without learning the deeper temporal context. This visualization reinforces the quantitative findings, demonstrating that while the baseline model is functional for trend following, it lacks the predictive horizon required for an effective inference system.

Figure 3.

Temporal Evolution of Mean Squared Error (MSE) during the 2025 Validation Period

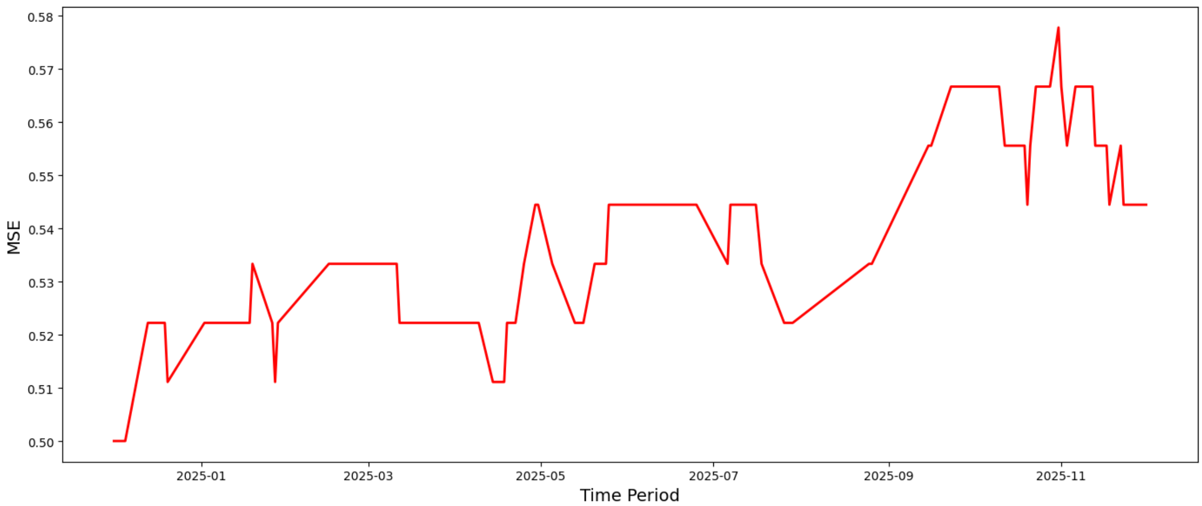

Figure 3 visualizes daily fluctuation of the Mean Squared Error (MSE) across the 2025 dataset. The distinct spikes in error magnitude correspond to periods of high market volatility, highlighting the Random Forest's limited capacity to generalize during non-stationary market regimes. This instability in error minimization further substantiates the requirement for recurrent architectures capable of maintaining long-term state dependencies.